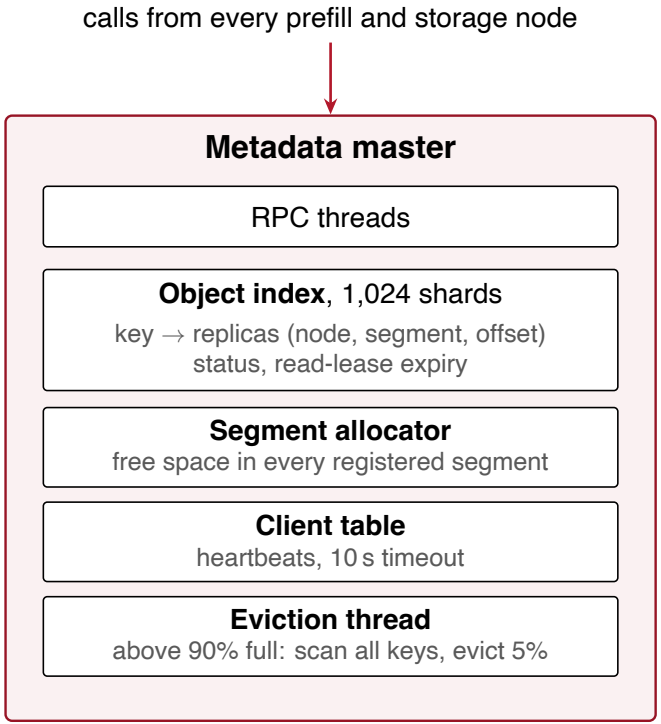
Mooncake keeps all KV-cache metadata in one master process

- **Index.** An in-memory map from every object's key to its replicas (node, segment, offset), status, and read-lease expiry: about 1 KB per key, split into 1,024 lock-protected shards.
- **Writes: two calls.** `PutStart` picks a segment and allocates space; the client copies the data by RDMA; `PutEnd` marks the object complete.
- **Lookups: two calls per cache hit.** `ExistKey`, then `GetReplicaList`, which returns the locations and grants a 10 s read lease.
- **Membership.** Storage nodes register their memory segments and send a heartbeat every second; a node silent for 10 s is dropped.
- **Eviction.** Above 90% pool use, a background thread scans every key and evicts 5% of objects, oldest lease first.



Defaults at Mooncake commit 292e839; memory per key measured in our runs. Optional HA: standby masters elected through etcd replay an operation log.